Lesson 24 material: Factorial Designs

9.1: Factorial Designs

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In Chapter 1 we briefly described a study conducted by Simone Schnall and her colleagues, in which they found that washing one's hands leads people to view moral transgressions as less wrong (Schnall, Benton, & Harvey, 2008)[1]. In a different but related study, Schnall and her colleagues investigated whether feeling physically disgusted causes people to make harsher moral judgments (Schnall, Haidt, Clore, & Jordan, 2008)[2]. In this experiment, they manipulated participants' feelings of disgust by testing them in either a clean room or a messy room that contained dirty dishes, an overflowing wastebasket, and a chewed-up pen. They also used a self-report questionnaire to measure the amount of attention that people pay to their own bodily sensations. They called this "private body consciousness." They measured their primary dependent variable, the harshness of people's moral judgments, by describing different behaviors (e.g., eating one's dead dog, failing to return a found wallet) and having participants rate the moral acceptability of each one on a scale of 1 to 7. Finally, the researchers asked participants to rate their current level of disgust and other emotions. The primary results of this study were that participants in the messy room were, in fact, more disgusted and made harsher moral judgments than participants in the clean room—but only if they scored relatively high in private body consciousness.

References

1. Schnall, S., Benton, J., & Harvey, S. (2008). With a clean conscience: Cleanliness reduces the severity of moral judgments. *Psychological Science, 19*(12), 1219-1222. doi: 10.1111/j.1467-9280.2008.02227.x ↵
2. Schnall, S., Haidt, J., Clore, G. L., & Jordan, A. H. (2008). Disgust as embodied moral judgment. *Personality and Social Psychology Bulletin, 34*, 1096–1109. ↵

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9.2: Setting Up a Factorial Experiment

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Learning Objectives

1. Explain why researchers often include multiple independent variables in their studies.
2. Define factorial design, and use a factorial design table to represent and interpret simple factorial designs.

Just as it is common for studies in psychology to include multiple levels of a single independent variable (placebo, new drug, old drug), it is also common for them to include multiple independent variables. Schnall and her colleagues studied the effect of both disgust and private body consciousness in the same study. Researchers' inclusion of multiple independent variables in one experiment is further illustrated by the following actual titles from various professional journals:

- The Effects of Temporal Delay and Orientation on Haptic Object Recognition
- Opening Closed Minds: The Combined Effects of Intergroup Contact and Need for Closure on Prejudice
- Effects of Expectancies and Coping on Pain-Induced Intentions to Smoke
- The Effect of Age and Divided Attention on Spontaneous Recognition
- The Effects of Reduced Food Size and Package Size on the Consumption Behavior of Restrained and Unrestrained Eaters

Just as including multiple levels of a single independent variable allows one to answer more sophisticated research questions, so too does including multiple independent variables in the same experiment. For example, instead of conducting one study on the effect of disgust on moral judgment and another on the effect of private body consciousness on moral judgment, Schnall and colleagues were able to conduct one study that addressed both questions. But including multiple independent variables also allows the researcher to answer questions about whether the effect of one independent variable depends on the level of another. This is referred to as an interaction between the independent variables. Schnall and her colleagues, for example, observed an interaction between disgust and private body consciousness because the effect of disgust depended on whether participants were high or low in private body consciousness. As we will see, interactions are often among the most interesting results in psychological research.

Factorial Designs

Overview

By far the most common approach to including multiple independent variables (which are often called factors) in an experiment is the factorial design. In a **factorial design**, each level of one independent variable is combined with each level of the others to produce all possible combinations. Each combination, then, becomes a condition in the experiment. Imagine, for example, an experiment on the effect of cell phone use (yes vs. no) and time of day (day vs. night) on driving ability. This is shown in the **factorial design table** in Figure . The columns of the table represent cell phone use, and the rows represent time of day. The four cells of the table represent the four possible combinations or conditions: using a cell phone during the day, not using a cell phone during the day, using a cell phone at night, and not using a cell phone at night. This particular design is referred to as a 2 × 2 (read "two-by-two") factorial design because it combines two variables, each of which has two levels.

If one of the independent variables had a third level (e.g., using a handheld cell phone, using a hands-free cell phone, and not using a cell phone), then it would be a 3 × 2 factorial design, and there would be six distinct conditions. Notice that the number of possible conditions is the product of the numbers of levels. A 2 × 2 factorial design has four conditions, a 3 × 2 factorial design has six conditions, a 4 × 5 factorial design would have 20 conditions, and so on. Also notice that each number in the notation represents one factor, one independent variable. So by looking at how many numbers are in the notation, you can determine how many independent variables there are in the experiment. 2 x 2, 3 x 3, and 2 x 3 designs all have two numbers in the notation and therefore all have two independent variables. The numerical value of each of the numbers represents the number of levels of each independent variable. A 2 means that the independent variable has two levels, a 3 means that the independent variable has three levels, a 4 means it has four levels, etc. To illustrate a 3 x 3 design has two independent variables, each with three levels, while a 2 x 2 x 2 design has three independent variables, each with two levels.

Figure: Factorial Design Table Representing a 2 × 2 Factorial Design

In principle, factorial designs can include any number of independent variables with any number of levels. For example, an experiment could include the type of psychotherapy (cognitive vs. behavioral), the length of the psychotherapy (2 weeks vs. 2 months), and the sex of the psychotherapist (female vs. male). This would be a 2 × 2 × 2 factorial design and would have eight conditions. Figure shows one way to represent this design. In practice, it is unusual for there to be more than three independent variables with more than two or three levels each. This is for at least two reasons: For one, the number of conditions can quickly become unmanageable. For example, adding a fourth independent variable with three levels (e.g., therapist experience: low vs. medium vs. high) to the current example would make it a 2 × 2 × 2 × 3 factorial design with 24 distinct conditions. Second, the number of participants required to populate all of these conditions (while maintaining a reasonable ability to detect a real underlying effect) can render the design unfeasible (for more information, see the discussion about the importance of adequate statistical power in Chapter 13). As a result, in the remainder of this section, we will focus on designs with two independent variables. The general principles discussed here extend in a straightforward way to more complex factorial designs.

Figure: Factorial Design Table Representing a 2 × 2 × 2 Factorial Design

Assigning Participants to Conditions

Recall that in a simple between-subjects design, each participant is tested in only one condition. In a simple within-subjects design, each participant is tested in all conditions. In a factorial experiment, the decision to take the between-subjects or within-subjects approach must be made separately for each independent variable. In a **between-subjects factorial design**, all of the independent variables are manipulated between subjects. For example, all participants could be tested either while using a cell phone *or* while not using a cell phone and either during the day *or* during the night. This would mean that each participant would be tested in one and only one condition. In a within-subjects factorial design, all of the independent variables are manipulated within subjects. All participants could be tested both while using a cell phone *and* while not using a cell phone and both during the day *and* during the night. This would mean that each participant would need to be tested in all four conditions. The advantages and disadvantages of these two approaches are the same as those discussed in Chapter 5. The between-subjects design is conceptually simpler, avoids order/carryover effects, and minimizes the time and effort of each participant. The within-subjects design is more efficient for the researcher and controls extraneous participant variables.

Since factorial designs have more than one independent variable, it is also possible to manipulate one independent variable between subjects and another within subjects. This is called a **mixed factorial design**. For example, a researcher might choose to treat cell phone use as a within-subjects factor by testing the same participants both while using a cell phone and while not using a cell phone (while counterbalancing the order of these two conditions). But they might choose to treat time of day as a between-subjects factor by testing each participant either during the day or during the night (perhaps because this only requires them to come in for testing once). Thus each participant in this mixed design would be tested in two of the four conditions.

Regardless of whether the design is between subjects, within subjects, or mixed, the actual assignment of participants to conditions or orders of conditions is typically done randomly.

Non-Manipulated Independent Variables

In many factorial designs, one of the independent variables is a **non-manipulated independent variable**. The researcher measures it but does not manipulate it. The study by Schnall and colleagues is a good example. One independent variable was disgust, which the researchers manipulated by testing participants in a clean room or a messy room. The other was private body consciousness, a participant variable which the researchers simply measured. Another example is a study by Halle Brown and colleagues in which participants were exposed to several words that they were later asked to recall (Brown, Kosslyn, Delamater, Fama, & Barsky, 1999)[1]. The manipulated independent variable was the type of word. Some were negative health-related words (e.g., *tumor, coronary*), and others were not health related (e.g., *election, geometry*). The non-manipulated independent variable was whether participants were high or low in hypochondriasis (excessive concern with ordinary bodily symptoms). The result of this study was that the participants high in hypochondriasis were better than those low in hypochondriasis at recalling the health-related words, but they were no better at recalling the non-health-related words.

Such studies are extremely common, and there are several points worth making about them. First, non-manipulated independent variables are usually participant variables (private body consciousness, hypochondriasis, self-esteem, gender, and so on), and as such, they are by definition between-subjects factors. For example, people are either low in hypochondriasis or high in hypochondriasis; they cannot be tested in both of these conditions. Second, such studies are generally considered to be experiments as long as at least one independent variable is manipulated, regardless of how many non-manipulated independent variables are included. Third, it is important to remember that causal conclusions can only be drawn about the manipulated independent variable. For example, Schnall and her colleagues were justified in concluding that disgust affected the harshness of their participants' moral judgments because they manipulated that variable and randomly assigned participants to the clean or messy room. But they would not have been justified in concluding that participants' private body consciousness affected the harshness of their participants' moral judgments because they did not manipulate that variable. It could be, for example, that having a strict moral code and a heightened awareness of one's body are both caused by some third variable (e.g., neuroticism). Thus it is important to be aware of which variables in a study are manipulated and which are not.

Non-Experimental Studies With Factorial Designs

Thus far we have seen that factorial experiments can include manipulated independent variables or a combination of manipulated and non-manipulated independent variables. But factorial designs can also include *only* non-manipulated independent variables, in which case they are no longer experiments but are instead non-experimental in nature. Consider a hypothetical study in which a researcher simply measures both the moods and the self-esteem of several participants—categorizing them as having either a positive or negative mood and as being either high or low in self-esteem—along with their willingness to have unprotected sexual intercourse. This can be conceptualized as a 2 × 2 factorial design with mood (positive vs. negative) and self-esteem (high vs. low) as non-manipulated between-subjects factors. Willingness to have unprotected sex is the dependent variable.

Again, because neither independent variable in this example was manipulated, it is a non-experimental study rather than an experiment. (The similar study by MacDonald and Martineau [2002][2] was an experiment because they manipulated their participants' moods.) This is important because, as always, one must be cautious about inferring causality from non-experimental studies because of the directionality and third-variable problems. For example, an effect of participants' moods on their willingness to have unprotected sex might be caused by any other variable that happens to be correlated with their moods.

References

1. Brown, H. D., Kosslyn, S. M., Delamater, B., Fama, A., & Barsky, A. J. (1999). Perceptual and memory biases for health-related information in hypochondriacal individuals. *Journal of Psychosomatic Research, 47*, 67–78. ↵
2. MacDonald, T. K., & Martineau, A. M. (2002). Self-esteem, mood, and intentions to use condoms: When does low self-esteem lead to risky health behaviors? *Journal of Experimental Social Psychology, 38*, 299–306. ↵

9.3: Interpreting the Results of a Factorial Experiment

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Learning Objectives

1. Distinguish between main effects and interactions, and recognize and give examples of each.
2. Sketch and interpret bar graphs and line graphs showing the results of studies with simple factorial designs.
3. Distinguish between main effects and simple effects, and recognize when an analysis of simple effects is required.

Graphing the Results of Factorial Experiments

The results of factorial experiments with two independent variables can be graphed by representing one independent variable on the *x*-axis and representing the other by using different colored bars or lines. (The *y*-axis is always reserved for the dependent variable.) Figure shows results for two hypothetical factorial experiments. The top panel shows the results of a 2 × 2 design. Time of day (day vs. night) is represented by different locations on the *x*-axis, and cell phone use (no vs. yes) is represented by different-colored bars. (It would also be possible to represent cell phone use on the *x*-axis and time of day as different-colored bars. The choice comes down to which way seems to communicate the results most clearly.) The bottom panel of Figure shows the results of a 4 × 2 design in which one of the variables is quantitative. This variable, psychotherapy length, is represented along the *x*-axis, and the other variable (psychotherapy type) is represented by differently formatted lines. This is a line graph rather than a bar graph because the variable on the x-axis is quantitative with a small number of distinct levels. Line graphs are also appropriate when representing measurements made over a time interval (also referred to as time series information) on the *x*-axis.

Figure: Two Ways to Plot the Results of a Factorial Experiment With Two Independent Variables

Main Effects

In factorial designs, there are three kinds of results that are of interest: main effects, interaction effects, and simple effects. A **main effect** is the effect of one independent variable on the dependent variable—averaging across the levels of the other independent variable. Thus there is one main effect to consider for each independent variable in the study. The top panel of Figure shows a main effect of cell phone use because driving performance was better, on average, when participants were not using cell phones than when they were. The blue bars are, on average, higher than the red bars. It also shows a main effect of time of day because driving performance was better during the day than during the night—both when participants were using cell phones and when they were not. Main effects are independent of each other in the sense that whether or not there is a main effect of one independent variable says nothing about whether or not there is a main effect of the other. The bottom panel of Figure , for example, shows a clear main effect of psychotherapy length. The longer the psychotherapy, the better it worked.

Interactions

There is an **interaction** effect (or just "interaction") when the effect of one independent variable depends on the level of another. Although this might seem complicated, you already have an intuitive understanding of interactions. As an everyday example, assume your friend asks you to go to a movie with another friend. Your response to her is, "well it depends on which movie you are going to see and who else is coming." You really want to see the big blockbuster summer hit but have little interest in seeing the cheesy romantic comedy. In other words, there is a main effect of type of movie on your decision. If your decision to go to see either of these movies further depends on who she is bringing with her then there is an interaction. For instance, if you will go to see the cheesy romantic comedy if she brings her hot friend you want to get to know better, but you will not go to this movie if she brings anyone else, then there is an interaction. Drug interactions are another good illustration of everyday interactions. Many older men take Viagara to assist them in the bedroom, and many men take nitrates to treat angina or chest pain. So each of these drugs is beneficial on its own (there are main effects of each on older men's well-being). But the combination of these two drugs can be lethal. In other words, there is a very important interaction between Viagara and heart medication that older men need to be aware of to prevent their untimely demise.

Let's now consider some examples of interactions from research. It probably would not surprise you to hear that the effect of receiving psychotherapy is stronger among people who are highly motivated to change than among people who are not motivated to change. This is an interaction because the effect of one independent variable (whether or not one receives psychotherapy) depends on the level of another (motivation to change). Schnall and her colleagues also demonstrated an interaction because the effect of whether the room was clean or messy on participants' moral judgments depended on whether the participants were low or high in private body consciousness. If they were high in private body consciousness, then those in the messy room made harsher judgments. If they were low in private body consciousness, then whether the room was clean or messy did not matter.

In many studies, the primary research question is about an interaction. The study by Brown and her colleagues was inspired by the idea that people with hypochondriasis are especially attentive to any negative health-related information. This led to the hypothesis that people high in hypochondriasis would recall negative health-related words more accurately than people low in hypochondriasis but recall non-health-related words about the same as people low in hypochondriasis. And of course, this is exactly what happened in this study.

Types of Interactions

The effect of one independent variable can depend on the level of the other in several different ways. First, there can be **spreading interactions**. Examples of spreading interactions are shown in the top two panels of Figure . In the top panel, independent variable "B" has an effect at level 1 of independent variable "A" (there is a difference in the height of the blue and red bars on the left side of the graph) but no effect at level 2 of independent variable "A." (there is no difference in the height of the blue and red bars on the right side of the graph). This is much like the study of Schnall and her colleagues where there was an effect of disgust for those high in private body consciousness but not for those low in private body consciousness. In the

middle panel, independent variable “B” has a stronger effect at level 1 of independent variable “A” than at level 2 (there is a larger difference in the height of the blue and red bars on the left side of the graph and a smaller difference in the height of the blue and red bars on the right side of the graph). This is like the hypothetical driving example where there was a strong effect of using a cell phone at night and a weaker effect of using a cell phone during the day. So to summarize, for spreading interactions there is an effect of one independent variable at one level of the other independent variable and there is either a weak effect or no effect of that independent variable at the other level of the other independent variable.

The second type of interaction that can be found is a **cross-over interaction.** A cross-over interaction is depicted in the bottom panel of Figure , independent variable “B” again has an effect at both levels of independent variable “A,” but the effects are in opposite directions. Another example of a crossover interaction comes from a study by Kathy Gilliland on the effect of caffeine on the verbal test scores of introverts and extraverts (Gilliland, 1980) [1]. Introverts perform better than extraverts when they have not ingested any caffeine. But extraverts perform better than introverts when they have ingested 4 mg of caffeine per kilogram of body weight.

Figure: Bar Graphs Showing Three Types of Interactions. In the top panel, one independent variable has an effect at one level of the second independent variable but not at the other. In the middle panel, one independent variable has a stronger effect at one level of the second independent variable than at the other. In the bottom panel, one independent variable has the opposite effect at one level of the second independent variable than at the other.

Figure shows examples of these same kinds of interactions when one of the independent variables is quantitative and the results are plotted in a line graph. Note that the top two figures depict the two kinds of spreading interactions that can be found while the bottom figure depicts a crossover interaction (the two lines literally “cross over” each other).

Figure: Line Graphs Showing Different Types of Interactions. In the top panel, one independent variable has an effect at one level of the second independent variable but not at the other. In the middle panel, one independent variable has a stronger effect at one level of the second independent variable than at the other. In the bottom panel, one independent variable has the opposite effect at one level of the second independent variable than at the other.

Simple Effects

When researchers find an interaction it suggests that the main effects may be a bit misleading. Think of the example of a crossover interaction where introverts were found to perform better on a test of verbal test performance than extraverts when they had not ingested any caffeine, but extraverts were found to perform better than introverts when they had ingested 4 mg of caffeine per kilogram of body weight. To examine the main effect of caffeine consumption, the researchers would have averaged across introversion and extraversion and simply looked at whether overall those who ingested caffeine had better or worse verbal memory test performance. Because the positive effect of caffeine on extraverts would be wiped out by the negative effects of caffeine on the introverts, no main effect of caffeine consumption would have been found. Similarly, to examine the main effect of personality, the researchers would have averaged across the levels of the caffeine variable to look at the effects of personality (introversion vs. extraversion) independent of caffeine. In this case, the positive effects extraversion in the caffeine condition would be wiped out by the negative effects of extraversion in the no caffeine condition. Does the absence of any main effects mean that there is no effect of caffeine and no effect of personality? No of course not. The presence of the interaction indicates that the story is more complicated, that the effects of caffeine on verbal test performance depend on personality. This is where simple effects come into play. **Simple effects** are a way of breaking down the interaction to figure out precisely what is going on. An interaction simply informs us that the effects of at least one independent variable depend on the level of another independent variable. Whenever an interaction is detected, researchers need to conduct additional analyses to determine where that interaction is coming from. Of course one may be able to visualize and interpret the interaction on a graph but a simple effects analysis provides researchers with a more sophisticated means of breaking down the interaction. Specifically, a simple effects analysis allows researchers to determine the effects of each independent variable at each level of the other independent variable. So while the researchers would average across the two levels of the personality variable to examine the effects of caffeine on verbal test performance in a main effects analysis, for a simple effects analysis the researchers would examine the effects of caffeine in introverts and then examine the effects of caffeine in extraverts. As we saw previously, the researchers also examined the effects of personality in the no caffeine condition and found that in this condition introverts performed better than extraverts. Finally, they examined the effects of personality in the caffeine condition and found that extraverts performed better than introverts in this condition. For a 2 x 2 design like this, there will be two main effects the researchers can explore and four simple effects.

Schnall and colleagues found a main effect of disgust on moral judgments (those in a messy room made harsher moral judgments). However, they also discovered an interaction between private body consciousness and disgust. In other words, the effect of disgust depended on private body consciousness. The presence of this interaction suggests the main effect may be a bit misleading. That is, it is not entirely accurate to say that those in a messy room made harsher moral judgments because this was only true for half of the participants. Using simple effects analyses, they were able to further demonstrate that for people high in private body consciousness, there was an effect of disgust on moral judgments. Further, they found that for those low in private body consciousness there was no effect of disgust on moral judgments. By examining the effect of disgust at each level of body consciousness using simple effects analyses, Schnall and colleagues were able to better understand the nature of the interaction.

As described previously, Brown and colleagues found an interaction between type of words (health related or not health related) and hypochondriasis (high or low) on word recall. To break down this interaction using simple effects analyses they examined the effect of hypochondriasis at each level of word type. Specifically, they examined the effect of hypochondriasis on recall of health-related words and then they subsequently examined the effect of hypochondriasis on recall of non-health related words. They found that people high in hypochondriasis were able to recall more health-related words than people low in hypochondriasis. In contrast, there was no effect of hypochondriasis on the recall of non-health related words.

Once again examining simple effects provides a means of breaking down the interaction and therefore it is only necessary to conduct these analyses when an interaction is present. When there is no interaction then the main effects will tell the complete and accurate story. To summarize, rather than averaging across the levels of the other independent variable, as is done in a main effects analysis, simple effects analyses are used to examine the effects of each independent variable at each level of the other independent variable(s). So a researcher using a 2×2 design with four conditions would need to look at 2 main effects and 4 simple effects. A researcher using a 2×3 design with six conditions would need to look at 2 main effects and 5 simple effects, while a researcher using a 3×3 design with nine conditions would need to look at 2 main effects and 6 simple effects. As you can see, while the number of main effects depends simply on the number of independent variables included (one main effect can be explored for each independent variable), the number of

simple effects analyses depends on the number of levels of the independent variables (because a separate analysis of each independent variable is conducted at each level of the other independent variable).

References

1. Gilliland, K. (1980). The interactive effect of introversion-extraversion with caffeine induced arousal on verbal performance. *Journal of Research in Personality, 14*, 482–492. ↵

9.4: Factorial Designs (Summary)

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Key Takeaways

- Researchers often include multiple independent variables in their experiments. The most common approach is the factorial design, in which each level of one independent variable is combined with each level of the others to create all possible conditions.
- Each independent variable can be manipulated between-subjects or within-subjects.
- Non-manipulated independent variables (gender) can be included in factorial designs, however, they limit the causal conclusions that can be made about the effects of the non-manipulated variable on the dependent variable.
- In a factorial design, the main effect of an independent variable is its overall effect averaged across all other independent variables. There is one main effect for each independent variable.
- There is an interaction between two independent variables when the effect of one depends on the level of the other. Some of the most interesting research questions and results in psychology are specifically about interactions.
- A simple effects analysis provides a means for researchers to break down interactions by examining the effect of each independent variable at each level of the other independent variable.

References

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Exercises

- Practice: Return to the five article titles presented at the beginning of this section. For each one, identify the independent variables and the dependent variable.
- Practice: Create a factorial design table for an experiment on the effects of room temperature and noise level on performance on the MCAT. Be sure to indicate whether each independent variable will be manipulated between-subjects or within-subjects and explain why.
- Practice: Sketch 8 different bar graphs to depict each of the following possible results in a 2 x 2 factorial experiment:
  - No main effect of A; no main effect of B; no interaction
  - Main effect of A; no main effect of B; no interaction
  - No main effect of A; main effect of B; no interaction
  - Main effect of A; main effect of B; no interaction
  - Main effect of A; main effect of B; interaction
  - Main effect of A; no main effect of B; interaction
  - No main effect of A; main effect of B; interaction
  - No main effect of A; no main effect of B; interaction